

Charges and Charging

Elias Xu

January 28, 2025

1 Charging Introduction

These are my notes on charges and charging for Physics!

Charges are fundamental properties for particles, just like **mass** and **spin**.

2 Fundamental Properties of Particles

There are three particles that matter for charges and charging.

Particle	Charge (in e)	Charge (in coulombs (C))
Proton	$1 e$	$1.6 \times 10^{-19} \text{ C}$
Electron	$- 1 e$	$1.6 \times 10^{-19} \text{ C}$
Neutron	$1 e$	$1.6 \times 10^{-19} \text{ C}$

Charges are usually sent in multiples of elementary charge, usually through electrons (because if you remove protons, the atoms themselves change).

Note: Elementary charges are done because they are the smallest stable charge. You can get quarks, but will reform to these particles.

3 Sending and Recieving Charge

There are three methods for sending and recieving charge:

- Friction
- Conduction
- Induction

3.1 Friction

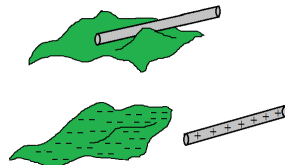


Figure 1: Example of Friction

When two objects rub with each other, the object with higher affinity will **take** electrons from the other. Usually, if you have two of the same object rub the same way, the objects have the same charge.

3.2 Conduction

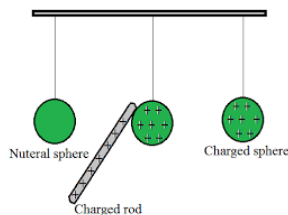


Figure 2: Example of Conduction

When two conductors (all objects are conductors in the end, if you try hard enough). If objects are identical, their charge will even out. Otherwise, they will just equalize.

3.3 Induction

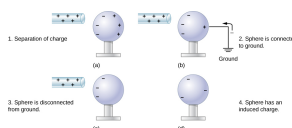


Figure 3: Example of Induction

A charge can change through proximity to an object. To equalize the charge afterwards, you can just tap the two objects together (if I remember correctly).

4 Polarized Neutral

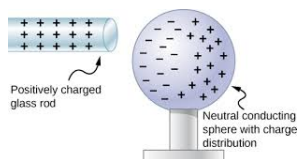


Figure 4: Example of a polarized neutral

A neutral object can be polarized through having the charged object attract the opposite charge of a neutral object, causing attraction through the distance.

5 Question Stuff

The question worksheet was *Electric Charge and Methods of Charging*, and the only weird thing was question 4. To solve, just do

$$\frac{1.5 \text{ m} \times (6.023 \times 10^{23} \text{ particles per m})}{1 \times 10^6 \text{ particles}} \times \frac{1.6 \times 10^{-19} \text{ C}}{e} = 0.144 \text{ C}$$

6.023×10^{23} is Avogadro's (Avocado's) number, which is equal to the number of particles per mole. One just has to divide by that the number of particles before one is charged, and then to convert elementary charges to coulombs.